

# SMAPhysics.com

## JEE Main Physics Practice Sheet — DPP-039

**Topic:** Hooke's Law, Spring Force, Spring Potential Energy & Equivalent Spring Systems

**Time Allowed:** 60 Minutes | **Max Marks:** 80 | **Marking Scheme:** +4, -1

- Q1.** A spring of spring constant  $k$  is stretched by an extension  $x$  from its natural length. The elastic potential energy stored in the spring is:
- (A)  $kx$   
 (B)  $\frac{1}{2}kx^2$   
 (C)  $kx^2$   
 (D)  $\frac{1}{4}kx^2$
- Q2.** Two identical springs of spring constant  $k$  are connected first in series and then in parallel. An external agent stretches both combinations by the same extension  $x$  from natural length. The ratio of elastic potential energy stored in series to that in parallel is:
- (A) 1 : 4  
 (B) 4 : 1  
 (C) 1 : 2  
 (D) 2 : 1
- Q3.** A block of mass  $m = 2$  kg moving at speed  $v = 4$  m/s on a frictionless horizontal floor collides with a light horizontal spring of spring constant  $k = 800$  N/m. The maximum compression of the spring is:
- (A) 0.10 m  
 (B) 0.30 m  
 (C) 0.20 m  
 (D) 0.40 m
- Q4.** A uniform spring of spring constant  $k$  and natural length  $L$  is cut into two pieces whose lengths are in the ratio 1 : 2. The spring constant of the shorter piece is:
- (A)  $\frac{k}{3}$   
 (B)  $\frac{3k}{2}$   
 (C)  $2k$   
 (D)  $3k$
- Q5.** A mass  $m$  is suspended gently from a vertical light spring of spring constant  $k$ . If the mass is released from rest at the position where the spring is unstretched, the maximum downward elongation of the spring is:
- (A)  $\frac{2mg}{k}$   
 (B)  $\frac{mg}{k}$   
 (C)  $\frac{mg}{2k}$   
 (D)  $\frac{4mg}{k}$
- Q6.** In Question 5, the equilibrium elongation of the spring when the oscillations die out completely is:
- (A)  $\frac{2mg}{k}$   
 (B)  $\frac{mg}{k}$   
 (C)  $\frac{mg}{2k}$   
 (D)  $\frac{4mg}{k}$
- Q7.** A spring is stretched by 2 cm and its potential energy is  $U$ . If it is stretched by an additional 4 cm (total extension 6 cm), the total potential energy stored in the spring becomes:
- (A)  $3U$   
 (B)  $6U$   
 (C)  $9U$   
 (D)  $4U$
- Q8.** A block of mass  $m$  is placed on a smooth horizontal table between two fixed walls and connected by two springs of stiffness  $k_1$  and  $k_2$  on opposite sides. When the block is displaced by  $x$  horizontally, the effective restoring force acting on the block is:
- (A)  $-(k_1 - k_2)x$   
 (B)  $-\left(\frac{k_1 k_2}{k_1 + k_2}\right)x$   
 (C)  $-(k_1 + k_2)x$   
 (D)  $-\frac{1}{2}(k_1 + k_2)x$
- Q9.** A non-Hookean spring has a restoring force given by  $F(x) = -kx^3$ . The potential energy  $U(x)$  stored in the spring when stretched by  $x$  from natural length ( $U(0) = 0$ ) is:
- (A)  $\frac{1}{4}kx^4$   
 (B)  $\frac{1}{3}kx^3$   
 (C)  $3kx^2$   
 (D)  $\frac{1}{2}kx^4$
- Q10.** Two springs  $A$  and  $B$  have spring constants  $k_A$  and  $k_B$  respectively, with  $k_A = 2k_B$ . If both springs are stretched by applying the same tensile pulling force  $F$ , the ratio of energy stored  $U_A/U_B$  is:
- (A) 1 : 2  
 (B) 2 : 1  
 (C) 1 : 4  
 (D) 4 : 1

- Q11.** A block of mass  $m = 1$  kg is dropped from a height  $h = 20$  cm onto a vertical spring ( $k = 400$  N/m). If  $x$  is the maximum compression of the spring, the value of  $x$  is ( $g = 10$  m/s<sup>2</sup>):
- (A) 0.20 m  
(B) 0.10 m  
(C) 0.125 m  
(D) 0.15 m
- Q12.** Two blocks of masses  $m_1$  and  $m_2$  connected by a light spring of stiffness  $k$  are pulled apart such that the spring is stretched by  $x_0$  and then released from rest on a frictionless floor. The maximum kinetic energy of mass  $m_1$  during subsequent motion is:
- (A)  $\frac{1}{2} \left( \frac{m_2}{m_1 + m_2} \right) kx_0^2$   
(B)  $\frac{1}{2} \left( \frac{m_1}{m_1 + m_2} \right) kx_0^2$   
(C)  $\frac{1}{2} kx_0^2$   
(D)  $\left( \frac{m_2}{m_1} \right) kx_0^2$
- Q13.** A spring-block system ( $m = 0.5$  kg,  $k = 200$  N/m) is compressed by  $x = 0.1$  m on a rough horizontal floor with  $\mu_k = 0.2$ . When released, the speed of the block as it passes through the unstretched position is ( $g = 10$  m/s<sup>2</sup>):
- (A) 1.0 m/s  
(B) 1.89 m/s  
(C) 1.50 m/s  
(D) 2.0 m/s
- Q14.** A light spring of stiffness  $k$  is compressed by  $x_1$  and then released. It pushes a ball of mass  $m$  horizontally off a smooth tabletop of height  $H$ . The horizontal range  $R$  of the ball before hitting the floor is:
- (A)  $x_1 \sqrt{\frac{2kH}{mg}}$   
(B)  $x_1 \sqrt{\frac{kH}{mg}}$   
(C)  $2x_1 \sqrt{\frac{kH}{mg}}$   
(D)  $\frac{x_1}{2} \sqrt{\frac{kH}{mg}}$
- Q15.** The potential energy curve  $U(x)$  of an ideal linear spring is a:
- (A) Straight line with positive slope  
(B) Symmetrical parabola opening upwards with vertex at  $x = 0$   
(C) Hyperbola with vertical asymptotes  
(D) Symmetrical parabola opening downwards
- Q16.** A block of mass  $M$  attached to a spring of stiffness  $k$  is moving with amplitude  $A$  on a smooth table. At the instant it passes through the equilibrium position, a lump of clay of mass  $m$  is gently dropped onto it and sticks. The new amplitude of oscillation  $A'$  is:
- (A)  $A \sqrt{\frac{M}{M+m}}$   
(B)  $A \sqrt{\frac{M+m}{M}}$   
(C)  $A \left( \frac{M}{M+m} \right)$   
(D)  $A \left( \frac{M+m}{M} \right)$
- Q17.** A spring balance reads  $F = 40$  N when suspended with an unknown mass. If the spring constant is  $k = 800$  N/m, the elastic potential energy stored in the balance spring is:
- (A) 2.0 J  
(B) 0.5 J  
(C) 1.0 J  
(D) 4.0 J
- Q18.** A block of mass  $m$  is attached to the lower end of a vertical spring of constant  $k$ . An external agent lowers the block slowly until it reaches its equilibrium position. The work done by the external agent during this process is:
- (A)  $-\frac{m^2 g^2}{2k}$   
(B)  $+\frac{m^2 g^2}{2k}$   
(C)  $-\frac{m^2 g^2}{k}$   
(D) Zero
- Q19.** A system consists of two massless springs of constants  $k_1$  and  $k_2$  connected in series. The force required to stretch the combination by total elongation  $X$  is:
- (A)  $(k_1 + k_2)X$   
(B)  $\left( \frac{k_1 k_2}{k_1 + k_2} \right) X$   
(C)  $\left( \frac{k_1 + k_2}{k_1 k_2} \right) X$   
(D)  $\frac{1}{2} (k_1 + k_2) X$
- Q20.** When a mass  $m$  attached to a spring of constant  $k$  performs simple harmonic motion with amplitude  $A$ , the ratio of its kinetic energy to potential energy at displacement  $x = \frac{A}{2}$  is:
- (A) 3 : 1  
(B) 1 : 3  
(C) 1 : 1  
(D) 2 : 1

# SMAPhysics.com — Answer Key & Detailed Solutions (DPP-039)

## Hooke's Law, Spring Force, Spring Potential Energy & Equivalent Spring Systems

| Q.No. | Ans | Q.No. | Ans | Q.No. | Ans | Q.No. | Ans |
|-------|-----|-------|-----|-------|-----|-------|-----|
| 1     | B   | 6     | B   | 11    | C   | 16    | A   |
| 2     | A   | 7     | C   | 12    | A   | 17    | C   |
| 3     | C   | 8     | C   | 13    | A   | 18    | A   |
| 4     | D   | 9     | A   | 14    | C   | 19    | B   |
| 5     | A   | 10    | A   | 15    | B   | 20    | A   |

### Detailed Step-by-Step Solutions

#### Sol 1. Option (B)

Work done by external agent against restoring force:

$$U = \int_0^x kx \, dx = \frac{1}{2}kx^2$$

#### Sol 2. Option (A)

In series:  $k_{\text{series}} = \frac{k}{2} \Rightarrow U_s = \frac{1}{2} \left( \frac{k}{2} \right) x^2 = \frac{1}{4}kx^2$ .

In parallel:  $k_{\text{parallel}} = 2k \Rightarrow U_p = \frac{1}{2}(2k)x^2 = kx^2$ .

$$\frac{U_s}{U_p} = \frac{\frac{1}{4}kx^2}{kx^2} = \frac{1}{4}$$

#### Sol 3. Option (C)

By conservation of mechanical energy ( $\Delta K + \Delta U = 0$ ):

$$\frac{1}{2}mv^2 = \frac{1}{2}kx_{\text{max}}^2 \Rightarrow x_{\text{max}} = v\sqrt{\frac{m}{k}}$$

$$x_{\text{max}} = 4\sqrt{\frac{2}{800}} = 4\sqrt{\frac{1}{400}} = \frac{4}{20} = 0.20 \text{ m}$$

#### Sol 4. Option (D)

For a uniform spring, spring constant is inversely proportional to natural length ( $k \propto 1/L$ ).

Shorter piece length  $L_1 = \frac{1}{3}L$ .

$$k_1 = \frac{kL}{L_1} = \frac{kL}{L/3} = 3k$$

#### Sol 5. Option (A)

From release ( $v_i = 0$ ) to maximum elongation  $x_m$  ( $v_f = 0$ ), mechanical energy is conserved:

$$mgx_m = \frac{1}{2}kx_m^2 \Rightarrow x_m = \frac{2mg}{k}$$

#### Sol 6. Option (B)

At equilibrium, upward spring force balances gravity:

$$kx_{\text{eq}} = mg \Rightarrow x_{\text{eq}} = \frac{mg}{k}$$

#### Sol 7. Option (C)

$$U = \frac{1}{2}kx_1^2 = \frac{1}{2}k(2)^2 = 2k.$$

When  $x_2 = 6 \text{ cm}$ :

$$U' = \frac{1}{2}k(6)^2 = 18k = 9(2k) = 9U$$

#### Sol 8. Option (C)

Displacing the block rightward by  $x$  compresses spring 2 by  $x$  (force  $k_2x$  to left) and stretches spring 1 by  $x$  (force  $k_1x$  to left).

$$F_{\text{net}} = -(k_1x + k_2x) = -(k_1 + k_2)x$$

#### Sol 9. Option (A)

Potential energy is defined by:

$$U(x) = - \int_0^x F_{\text{spring}} \, dx = - \int_0^x (-kx^3) \, dx = \frac{1}{4}kx^4$$

#### Sol 10. Option (A)

Potential energy in terms of tension force:  $U = \frac{F^2}{2k}$ .

For the same tensile force  $F$ :

$$\frac{U_A}{U_B} = \frac{k_B}{k_A} = \frac{k_B}{2k_B} = \frac{1}{2}$$

#### Sol 11. Option (C)

From drop to maximum compression  $x$ :

$$mg(h+x) = \frac{1}{2}kx^2 \Rightarrow 1(10)(0.2+x) = \frac{1}{2}(400)x^2$$

$$2 + 10x = 200x^2 \Rightarrow 100x^2 - 5x - 1 = 0$$

$$x = \frac{5 + \sqrt{25 - 4(100)(-1)}}{200} = \frac{5 + \sqrt{425}}{200} \approx \frac{5 + 20.6}{200} \approx 0.128 \text{ m} \approx 0$$

#### Sol 12. Option (A)

In the center-of-mass frame, momentum conservation gives

$$m_1v_1 = m_2v_2 \Rightarrow v_2 = \frac{m_1}{m_2}v_1.$$

Total energy:

$$K_1 + K_2 = \frac{1}{2}kx_0^2 \Rightarrow K_1 \left( 1 + \frac{m_1}{m_2} \right) = \frac{1}{2}kx_0^2$$

$$K_1 = \frac{1}{2} \left( \frac{m_2}{m_1 + m_2} \right) kx_0^2$$

#### Sol 13. Option (A)

By Work-Energy Theorem:

$$W_{\text{spring}} + W_{\text{friction}} = \Delta K$$

$$\frac{1}{2}kx^2 - \mu_k mgx = \frac{1}{2}mv^2$$

$$\frac{1}{2}(200)(0.1)^2 - 0.2(0.5)(10)(0.1) = \frac{1}{2}(0.5)v^2$$

$$1.0 - 0.1 = 0.25v^2 \Rightarrow 0.9 = 0.25v^2 \Rightarrow v^2 = 3.6 \Rightarrow v \approx 1.89 \text{ m/s}$$

**Sol 14. Option (C)**

Speed on leaving table:  $\frac{1}{2}mv^2 = \frac{1}{2}kx_1^2 \Rightarrow v = x_1\sqrt{\frac{k}{m}}$ .

Time of flight:  $t = \sqrt{\frac{2H}{g}}$ .

Range:

$$R = vt = x_1\sqrt{\frac{k}{m}}\sqrt{\frac{2H}{g}} = x_1\sqrt{\frac{2kH}{mg}}$$

**Sol 15. Option (B)**

$U(x) = \frac{1}{2}kx^2$  is an upward-opening parabola with its minimum vertex at  $x = 0$ .

**Sol 16. Option (A)**

At equilibrium, speed before drop is  $v_0 = \omega A = \sqrt{\frac{k}{M}}A$ .

Momentum is conserved when clay drops:

$$Mv_0 = (M + m)v' \Rightarrow v' = \frac{M}{M + m}v_0$$

New amplitude:

$$A' = \frac{v'}{\omega'} = \left(\frac{Mv_0}{M + m}\right)\sqrt{\frac{M + m}{k}} = A\sqrt{\frac{M}{M + m}}$$

**Sol 17. Option (C)**

Energy stored:

$$U = \frac{F^2}{2k} = \frac{40^2}{2(800)} = \frac{1600}{1600} = 1.0 \text{ J}$$

**Sol 18. Option (A)**

Equilibrium position:  $x_0 = \frac{mg}{k}$ .

$\Delta K = 0 \Rightarrow W_g + W_s + W_{\text{ext}} = 0$ .

$$mgx_0 - \frac{1}{2}kx_0^2 + W_{\text{ext}} = 0 \Rightarrow \frac{m^2g^2}{k} - \frac{m^2g^2}{2k} + W_{\text{ext}} = 0$$

$$W_{\text{ext}} = -\frac{m^2g^2}{2k}$$

**Sol 19. Option (B)**

For springs in series, the equivalent spring constant is

$$k_{\text{eq}} = \frac{k_1k_2}{k_1 + k_2}.$$

Restoring force:

$$F = k_{\text{eq}}X = \left(\frac{k_1k_2}{k_1 + k_2}\right)X$$

**Sol 20. Option (A)**

At  $x = A/2$ :

$$U = \frac{1}{2}kx^2 = \frac{1}{2}k\left(\frac{A}{2}\right)^2 = \frac{1}{8}kA^2$$

$$K = E - U = \frac{1}{2}kA^2 - \frac{1}{8}kA^2 = \frac{3}{8}kA^2$$

$$\frac{K}{U} = \frac{3/8}{1/8} = \frac{3}{1}$$